Two networks.
One control plane.

Fixed access and mobile access are managed by two disjoint toolchains, two data models and two on-call rotations. Aether puts them behind one.

10 PROTOCOLS · ONE BINARY · OPEN AGENT · MANAGED PLATFORM

WHO BUILDS THIS

Not a first attempt.
A replacement.

Aether is the platform we built to replace the seven stacks underneath a fleet we already run.

5M+	7	10
devices under management	firmware platforms we build on	protocols in one binary

Firmware development is our other business — OpenWRT, OpenWiFi, wlan-ap, RDK-B and QSDK, plus camera and IoT firmware on embedded Linux. The protocol coverage looks the way it does because we have had to live with all of it.

THE PROBLEM

The access network is managed twice.

Every operator runs both of these. Almost nobody runs them from one place.

CPE / WiFi

· uCentral · TIP OpenWiFi
· TR-369 / USP
· WRP · RDK-B
· IEEE 1905.1 EasyMesh
· OpenSync

BROADBAND TEAM

Transport / RAN

· NETCONF + YANG
· SNMP v1/v2c/v3
· gNMI streaming
· O-RAN A1 policy
· VES event collector

RAN + TRANSPORT TEAM

When a subscriber says “my connection is bad” —

no single system can see the home, the backhaul and the radio.

PROTOCOL COVERAGE

Ten protocols.
One binary.

uCentral / TIP	WebSocket + JSON · OpenWiFi, OpenWrt
TR-369 / USP	WebSocket + MQTT · protobuf · QSDK, prplOS
WRP / Xmidt	RDK-B, zero-install via Parodus
IEEE 1905.1	EtherType 0x893a · CMDU · EasyMesh R1–R5
OpenSync	MQTT · Plume migration path
MQTT	3.1 / 5.0 clustered broker · USP MTP
NETCONF / YANG	SSH + XML · RFC 6241 / 6242
SNMP	v1 / v2c / v3 · polling + traps
gNMI	gRPC + protobuf · OpenConfig telemetry
O-RAN A1 / VES	A1 policy → Near-RT RIC · VES collector

ABOVE THE LINE: CPE.   BELOW: TRANSPORT, BACKHAUL AND RAN.

PRECISION

Aether is not a RIC.

Here is what it is.

IS

- A1 — deliver, withdraw, fetch and status policy
- O1 — get-config / edit-config over real NETCONF
- VES 7 collector — events into ONAP-style OSS
- Device.Cellular — IMEI, IMSI, RSRP, RSRQ, SINR on FWA CPE
- Federated ML — FedAvg aggregation across the fleet

ON THE ROADMAP

- Non-RT RIC and an rApp runtime over R1
- IoT device management — LwM2M, ONVIF, MQTT ingest
- STOMP and CoAP in the OpenWrt agent

IS NOT

A Near-RT RIC. No E2 termination, and so no xApp runtime — that belongs at the edge on a sub-second control loop, not in a cloud control plane. Different product.

The hard part was never A1. It was a data model where a TR-181 parameter from an OpenWrt box and a gNMI subscription from an aggregation switch are both first-class.

OPEN SOURCE · OPENWRT

ac-client.

Download. Install. Sign up.

The OpenWrt USP agent is open source under BSD 3-Clause. Three commands and the router is under management. QSDK, prplOS and RDK-B run their own native agents — nothing of ours to install there.

```
opkg install ac-client luci-app-aclient
uci set optimacs.@config[0].claim_token='...'
/etc/init.d/ac-client start
```

TR-369 / USP 1.3	TP-469 conformance tested
Post-quantum mTLS	X25519 + ML-KEM-768, always on
UCI-backed TR-181	47 backend operations
WiFi 7 + Cellular	EHT modes and Device.Cellular

1,500+ OpenWrt router models. No custom firmware image.

The OpenWrt package is open. The platform is a managed service.

DEPLOYMENT

Meet the device where it is.

No reflash. No certification cycle. No truck roll.

- **RDK-B**
  Terminates the Parodus/WRP session already running.
  No firmware change.
- **OpenWrt**
  opkg package install. 30k+ router models.
- **QSDK**
  Agent wrapping cfg80211tool and qca-hostapd.
- **prplOS**
  Speaks the TR-181 data model bus already present.
- **OpenWiFi**
  Native uCentral client.
- **Aggregation gear**
  NETCONF or gNMI. SNMP for anything older.

Config changes ship as JSON Patch (RFC 6902)

~200 bytes per update, not a ~50 KB full-config push.

OEM / ODM

We write the firmware as well.

Agent implementation, platform HALs and full firmware builds. An engagement, priced per device.

OpenWRT · OpenWiFi · wlan-ap
the stacks most shops stop at

RDK-B
where an ODM assumes they are stuck with the silicon vendor

QSDK
QCA HAL, chipset data model

Camera / embedded Linux
RTSP, ONVIF

IoT / embedded
constrained devices, custom silicon

Firmware, agent and platform from one team.

One accountability line instead of four.

Ten protocols.
One control plane.

Built for a messy access network. Multiple CPE silicon vendors, a platform inherited from an acquisition, aggregation gear still on SNMP, and fixed wireless CPE nobody has a dashboard for.

— $0.30 / device / month at volume — no per-device AI tax
— Open source OpenWrt agent, managed platform, EU / US residency
— Self-hosting licensed on Enterprise

aether-io.com

Running today · aether-io.com/pricing

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